

PROJECT OF DATABASE SYSTEMS

PHASE II

SUBMITTED BY:

HUZAIFA AJMAL, UMER RAUF

SAP ID:

70142166, 70150397

use project

```
create table Users (  
  UserID int primary key,  
  FirstName varchar(50),  
  LastName varchar(50),  
  Email varchar(100) UNIQUE,  
  RegistrationDate date,  
  DateOfBirth date,  
  Address_Street varchar(100),  
  Address_City varchar(50),  
  Address_ZipCode varchar(10),  
  UserType varchar(20),  
  CHECK (UserType IN ('citizen', 'collector', 'admin'))  
  
  create table UserPhoneNumbers (  
    UserID int,  
    PhoneNumber varchar(20),  
    PRIMARY KEY (UserID, PhoneNumber),  
    foreign key (UserID) references Users(UserID))  
  
  create table Citizens (  
    CitizenID int primary key,  
    UserID int UNIQUE,  
    PreferredPickupTime TIME,  
    NumberOfRequestsMade int,  
    Location_Latitude decimal(9,6),  
    Location_Longitude decimal(9,6),  
    LastRequestDate date,  
    foreign key (UserID) references Users(UserID))  
  
  create table CitizenFeedbacks (  
    CitizenID int,  
    Feedback varchar(100),
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PRIMARY KEY (CitizenID, Feedback),
foreign key (CitizenID) references Citizens(CitizenID))

create table Collectors (
CollectorID int primary key,
UserID int UNIQUE,
LicenseNumber varchar(50),
CurrentVehicleID int,
TotalCollections int,
Shift_Start TIME,
Shift_End TIME,
EmploymentDate date,
ActiveStatus bit,
AreaAssigned varchar(100),
foreign key (UserID) references Users(UserID))

create table CollectorAssignedRequests (
CollectorID int,
RequestID int,
PRIMARY KEY (CollectorID, RequestID),
foreign key (CollectorID) references Collectors(CollectorID))

create table Admins (
AdminID int primary key,
UserID int UNIQUE,
ReportsGenerated int,
LastLoginDate date,
WorkShift varchar(50),
ApprovalCount int,
foreign key (UserID) references Users(UserID))

create table AdminManagedZones (
AdminID int,
ZoneName varchar(100),
PRIMARY KEY (AdminID, ZoneName),
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foreign key (AdminID) references Admins(AdminID))

create table PickupRequests (
RequestID int primary key,
CitizenID int,
AdminID int,
CollectorID int,
RequestDate date,
ApprovalStatus varchar(20),
TotalWeight decimal(6,2),
Pickup_Street varchar(100),
Pickup_City varchar(50),
Pickup_ZipCode varchar(10),
PickupStatus varchar(20),
foreign key (CitizenID) references Citizens(CitizenID),
foreign key(AdminID) references Admins(AdminID),
foreign key (CollectorID) references Collectors(CollectorID),
CHECK (ApprovalStatus IN ('Pending', 'Approved', 'Rejected')),
CHECK (PickupStatus IN ('Scheduled', 'Collected', 'Missed')))

create table PickupRequest_WasteItems (
RequestID int,
WasteItem varchar(100),
PRIMARY KEY (RequestID, WasteItem),
foreign key (RequestID) references PickupRequests(RequestID))

create table Vehicles (
VehicleID int primary key,
VehicleType varchar(50),
Capacity decimal(6,2),
AssignedCollectorID int UNIQUE,
Location_Latitude decimal(9,6),
Location_Longitude decimal(9,6),
Status varchar(20),
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FuelType varchar(30),
OdometerReading int,
foreign key (AssignedCollectorID) references Collectors(CollectorID),
CHECK (Status IN ('Available', 'Assigned', 'Maintenance'))
create table RecycleCenters (
CenterID int primary key,
CenterName varchar(100),
Address_Street varchar(100),
Address_City varchar(50),
Address_ZipCode varchar(10),
ContactNumber varchar(20),
AdminID int,
TotalWasteReceived decimal(10,2),
CenterCapacity decimal(10,2),
InventoryID int UNIQUE,
EstablishedDate date,
foreign key (AdminID) references Admins(AdminID),
foreign key (InventoryID) references Inventory(InventoryID))
create table RecycleCenter_WasteCategories (
CenterID int,
WasteCategory varchar(50),
PRIMARY KEY (CenterID, WasteCategory),
foreign key (CenterID) references RecycleCenters(CenterID))
create table Recyclers (
RecyclerID int primary key,
CenterID int,
FirstName varchar(50),
LastName varchar(50),
Email varchar(100),
Phone varchar(20),
Specialization varchar(100),

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Shift varchar(50),
foreign key (CenterID) references RecycleCenters(CenterID))
create table Recycler_RecycledItemsLog (
RecyclerID int,
Item varchar(100),
PRIMARY KEY (RecyclerID, Item),
foreign key (RecyclerID) references Recyclers(RecyclerID))
create table Inventory (
InventoryID int primary key,
CenterID int UNIQUE,
MaxCapacity decimal(10,2),
LastUpdatedDate date,
TotalWeightStored decimal(10,2),
InventoryManagerID int)
CREATE TABLE Inventory_ItemTypes (
InventoryID int,
ItemType varchar(50),
PRIMARY KEY (InventoryID, ItemType),
foreign key (InventoryID) references Inventory(InventoryID))
create table Inventory_StockLevels (
InventoryID int,
ItemType varchar(50),
Quantity decimal(8,2),
PRIMARY KEY (InventoryID, ItemType),
foreign key (InventoryID, ItemType) references Inventory_ItemTypes(InventoryID, ItemType))
create table Reports (
ReportID int primary key,
AdminID int,
GeneratedDate date,
TimePeriodCovered varchar(50),
TotalPickups int,

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TotalRecycled int,
IssuesFound text,
WasteTypeBreakdown text,
ReviewedBy varchar(100),
LocationStats_Latitude DECIMAL(9,6),
LocationStats_Longitude DECIMAL(9,6),
foreign key (AdminID) references Admins(AdminID))
create table Report_TopCollectors (
ReportID int,
CollectorID int,
PRIMARY KEY (ReportID, CollectorID),
foreign key (ReportID) references Reports(ReportID),
foreign key (CollectorID) references Collectors(CollectorID))
create table Collector_Deliveries (
CollectorID int,
CenterID int,
DeliveryDate date,
PRIMARY KEY (CollectorID, CenterID, DeliveryDate),
foreign key (CollectorID) references Collectors(CollectorID),
foreign key (CenterID) references RecycleCenters(CenterID))

INSERT INTO Users
VALUES
(1, 'John', 'Doe', 'john.doe@example.com', '2022-01-15', '1985-05-10', '123 Main St', 'Springfield', '12345',
'citizen'),
(2, 'Jane', 'Smith', 'jane.smith@example.com', '2022-02-20', '1990-08-25', '456 Oak Ave', 'Shelbyville', '23456',
'citizen'),
(3, 'Robert', 'Brown', 'robert.brown@example.com', '2022-03-05', '1978-11-30', '789 Pine Rd', 'Capital City',
'34567', 'citizen'),
(4, 'Emily', 'Davis', 'emily.davis@example.com', '2022-04-10', '1995-04-15', '101 Elm Blvd', 'Ogdenville',
'45678', 'citizen'),
(5, 'Michael', 'Wilson', 'michael.wilson@example.com', '2022-05-25', '1982-07-20', '202 Maple Dr', 'North
Haverbrook', '56789', 'citizen'),

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(6, 'Sarah', 'Taylor', 'sarah.taylor@example.com', '2022-06-30', '1993-09-12', '303 Cedar Ln', 'Brockway',
'67890', 'citizen'),

(7, 'David', 'Anderson', 'david.anderson@example.com', '2022-07-15', '1975-12-05', '404 Birch St',
'Springfield', '78901', 'citizen'),

(8, 'Jessica', 'Thomas', 'jessica.thomas@example.com', '2022-08-20', '1988-03-18', '505 Spruce Ave',
'Shelbyville', '89012', 'citizen'),

(9, 'Daniel', 'Jackson', 'daniel.jackson@example.com', '2022-09-25', '1991-06-22', '606 Willow Rd', 'Capital
City', '90123', 'citizen'),

(10, 'Lisa', 'White', 'lisa.white@example.com', '2022-10-30', '1980-01-08', '707 Aspen Blvd', 'Ogdenville',
'01234', 'citizen'),

(11, 'Matthew', 'Harris', 'matthew.harris@example.com', '2022-11-05', '1972-10-14', '808 Redwood Dr',
'North Haverbrook', '12340', 'citizen'),

(12, 'Amanda', 'Martin', 'amanda.martin@example.com', '2022-12-10', '1994-02-28', '909 Sequoia Ln',
'Brockway', '23451', 'citizen'),

(13, 'Christopher', 'Garcia', 'christopher.garcia@example.com', '2023-01-15', '1987-07-03', '111 Fir St',
'Springfield', '34562', 'citizen'),

(14, 'Ashley', 'Lee', 'ashley.lee@example.com', '2023-02-20', '1996-04-17', '222 Juniper Ave', 'Shelbyville',
'45673', 'citizen'),

(15, 'James', 'Clark', 'james.clark@example.com', '2023-03-25', '1979-08-09', '333 Magnolia Rd', 'Capital City',
'56784', 'citizen');

INSERT INTO Users

VALUES

(16, 'Kevin', 'Adams', 'kevin.adams@example.com', '2021-05-10', '1983-09-15', '444 Oak St', 'Ogdenville',
'67895', 'collector'),

(17, 'Nicole', 'Baker', 'nicole.baker@example.com', '2021-06-15', '1990-12-20', '555 Pine Ave', 'North
Haverbrook', '78906', 'collector'),

(18, 'Brian', 'Carter', 'brian.carter@example.com', '2021-07-20', '1976-03-25', '666 Maple Rd', 'Brockway',
'89017', 'collector'),

(19, 'Rachel', 'Evans', 'rachel.evans@example.com', '2021-08-25', '1985-06-30', '777 Cedar Blvd', 'Springfield',
'90128', 'collector'),

(20, 'Steven', 'Foster', 'steven.foster@example.com', '2021-09-30', '1992-01-05', '888 Birch Dr', 'Shelbyville',
'01239', 'collector'),

(21, 'Melissa', 'Green', 'melissa.green@example.com', '2021-10-05', '1988-04-10', '999 Spruce Ln', 'Capital
City', '12350', 'collector'),

(22, 'Timothy', 'Hill', 'timothy.hill@example.com', '2021-11-10', '1974-07-15', '1010 Willow St', 'Ogdenville',
'23461', 'collector'),

(23, 'Laura', 'Ingram', 'laura.ingram@example.com', '2021-12-15', '1993-10-20', '1111 Aspen Ave', 'North
Haverbrook', '34572', 'collector'),

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(24, 'Jason', 'King', 'jason.king@example.com', '2022-01-20', '1981-02-25', '1212 Redwood Rd', 'Brockway', '45683', 'collector'),

(25, 'Stephanie', 'Lopez', 'stephanie.lopez@example.com', '2022-02-25', '1995-05-30', '1313 Sequoia Blvd', 'Springfield', '56794', 'collector'),

(26, 'Ryan', 'Mitchell', 'ryan.mitchell@example.com', '2022-03-30', '1978-08-05', '1414 Fir Dr', 'Shelbyville', '67805', 'collector'),

(27, 'Kimberly', 'Nelson', 'kimberly.nelson@example.com', '2022-04-05', '1986-11-10', '1515 Juniper Ln', 'Capital City', '78916', 'collector'),

(28, 'Eric', 'Owens', 'eric.owens@example.com', '2022-05-10', '1991-01-15', '1616 Magnolia St', 'Ogdenville', '89027', 'collector'),

(29, 'Angela', 'Perez', 'angela.perez@example.com', '2022-06-15', '1973-04-20', '1717 Oak Ave', 'North Haverbrook', '90138', 'collector'),

(30, 'Justin', 'Roberts', 'justin.roberts@example.com', '2022-07-20', '1989-07-25', '1818 Pine Rd', 'Brockway', '01249', 'collector');

INSERT INTO Users

VALUES

(31, 'Brandon', 'Scott', 'brandon.scott@example.com', '2020-01-05', '1980-10-12', '1919 Cedar Blvd', 'Springfield', '12350', 'admin'),

(32, 'Megan', 'Turner', 'megan.turner@example.com', '2020-02-10', '1994-03-17', '2020 Birch Dr', 'Shelbyville', '23461', 'admin'),

(33, 'Nathan', 'Walker', 'nathan.walker@example.com', '2020-03-15', '1977-06-22', '2121 Spruce Ln', 'Capital City', '34572', 'admin'),

(34, 'Olivia', 'Young', 'olivia.young@example.com', '2020-04-20', '1984-09-27', '2222 Willow St', 'Ogdenville', '45683', 'admin'),

(35, 'Patrick', 'Zimmerman', 'patrick.zimmerman@example.com', '2020-05-25', '1997-12-02', '2323 Aspen Ave', 'North Haverbrook', '56794', 'admin'),

(36, 'Hannah', 'Allen', 'hannah.allen@example.com', '2020-06-30', '1982-02-07', '2424 Redwood Rd', 'Brockway', '67805', 'admin'),

(37, 'Alexander', 'Bennett', 'alexander.bennett@example.com', '2020-07-05', '1975-05-12', '2525 Sequoia Blvd', 'Springfield', '78916', 'admin'),

(38, 'Victoria', 'Coleman', 'victoria.coleman@example.com', '2020-08-10', '1990-08-17', '2626 Fir Dr', 'Shelbyville', '89027', 'admin'),

(39, 'Samuel', 'Dixon', 'samuel.dixon@example.com', '2020-09-15', '1987-11-22', '2727 Juniper Ln', 'Capital City', '90138', 'admin'),

(40, 'Christina', 'Edwards', 'christina.edwards@example.com', '2020-10-20', '1979-01-27', '2828 Magnolia St', 'Ogdenville', '01249', 'admin'),

(41, 'Benjamin', 'Flores', 'benjamin.flores@example.com', '2020-11-25', '1993-04-02', '2929 Oak Ave', 'North Haverbrook', '12350', 'admin'),

(42, 'Samantha', 'Gonzalez', 'samantha.gonzalez@example.com', '2020-12-30', '1986-07-07', '3030 Pine Rd', 'Brockway', '23461', 'admin'),

(43, 'Dylan', 'Hughes', 'dylan.hughes@example.com', '2021-01-05', '1972-10-12', '3131 Cedar Blvd', 'Springfield', '34572', 'admin'),

(44, 'Natalie', 'Irwin', 'natalie.irwin@example.com', '2021-02-10', '1995-01-17', '3232 Birch Dr', 'Shelbyville', '45683', 'admin'),

(45, 'Caleb', 'Jenkins', 'caleb.jenkins@example.com', '2021-03-15', '1981-04-22', '3333 Spruce Ln', 'Capital City', '56794', 'admin');

select * from users



ID	FirstName	LastName	Email	RegistrationDate	DateOfBirth	Address_Street	Address_City	Address_ZipCode	UserType	LastLogin	Account
1	John	Doe	john.doe@example.com	2022-01-15	1985-05-10	123 Main St	Springfield	12345	citizen	2022-01-15	Active
2	Jane	Smith	jane.smith@example.com	2022-02-20	1990-08-25	456 Oak Ave	Shelbyville	23456	citizen	2022-02-20	Active
3	Robert	Brown	robert.brown@example.com	2022-03-05	1978-11-30	789 Pine Rd	Capital City	34567	citizen	2022-03-05	Active
4	Emily	Davis	emily.davis@example.com	2022-04-10	1995-04-15	101 Elm Blvd	Ogdenville	45678	citizen	2022-04-10	Active
5	Michael	Wilson	michael.wilson@example.com	2022-05-25	1982-07-20	202 Maple Dr	North Haverbrook	56789	citizen	2022-05-25	Active
6	Sarah	Taylor	sarah.taylor@example.com	2022-06-30	1993-09-12	303 Cedar Ln	Brockway	67890	citizen	2022-06-30	Active

INSERT INTO Citizens (CitizenID, UserID, PreferredPickupTime, NumberOfRequestsMade, Location_Latitude, Location_Longitude, LastRequestDate)

VALUES

(1, 1, '08:00:00', 5, 34.052235, -118.243683, '2023-05-10'),

(2, 2, '09:30:00', 3, 34.052236, -118.243684, '2023-05-12'),

(3, 3, '10:00:00', 7, 34.052237, -118.243685, '2023-05-15'),

(4, 4, '11:00:00', 2, 34.052238, -118.243686, '2023-05-18'),

(5, 5, '13:00:00', 4, 34.052239, -118.243687, '2023-05-20'),

(6, 6, '14:00:00', 6, 34.052240, -118.243688, '2023-05-22'),

(7, 7, '15:30:00', 1, 34.052241, -118.243689, '2023-05-25'),

(8, 8, '16:00:00', 8, 34.052242, -118.243690, '2023-05-28'),

(9, 9, '07:30:00', 5, 34.052243, -118.243691, '2023-06-01'),

(10, 10, '12:00:00', 3, 34.052244, -118.243692, '2023-06-05'),

(11, 11, '08:30:00', 9, 34.052245, -118.243693, '2023-06-10'),

(12, 12, '10:30:00', 2, 34.052246, -118.243694, '2023-06-15'),

(13, 13, '09:00:00', 4, 34.052247, -118.243695, '2023-06-20'),

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(14, 14, '11:30:00', 7, 34.052248, -118.243696, '2023-06-25'),
(15, 15, '14:30:00', 6, 34.052249, -118.243697, '2023-06-30');
select* from citizens
```

CitizenID	UserID	NumberOfRequestsMade	Location_Longitude	LastRequestDate
1	1	5	-118.243683	2023-05-10
2	2	3	-118.243684	2023-05-12
3	3	7	-118.243685	2023-05-15
4	4	2	-118.243686	2023-05-18
5	5	4	-118.243687	2023-05-20
6	6	6	-118.243688	2023-05-22

```
INSERT INTO Collectors (CollectorID, UserID, LicenseNumber, CurrentVehicleID, TotalCollections, Shift_Start,
Shift_End, EmploymentDate, ActiveStatus, AreaAssigned)
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VALUES
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(1, 16, 'LIC-1001', 1, 120, '07:00:00', '15:00:00', '2021-05-10', 1, 'Springfield'),
(2, 17, 'LIC-1002', 2, 95, '08:00:00', '16:00:00', '2021-06-15', 1, 'Shelbyville'),
(3, 18, 'LIC-1003', 3, 80, '09:00:00', '17:00:00', '2021-07-20', 1, 'Capital City'),
(4, 19, 'LIC-1004', 4, 110, '10:00:00', '18:00:00', '2021-08-25', 1, 'Ogdenville'),
(5, 20, 'LIC-1005', 5, 65, '11:00:00', '19:00:00', '2021-09-30', 1, 'North Haverbrook'),
(6, 21, 'LIC-1006', 6, 150, '06:00:00', '14:00:00', '2021-10-05', 1, 'Brockway'),
(7, 22, 'LIC-1007', 7, 70, '12:00:00', '20:00:00', '2021-11-10', 1, 'Springfield'),
(8, 23, 'LIC-1008', 8, 85, '07:30:00', '15:30:00', '2021-12-15', 1, 'Shelbyville'),
(9, 24, 'LIC-1009', 9, 100, '08:30:00', '16:30:00', '2022-01-20', 1, 'Capital City'),
(10, 25, 'LIC-1010', 10, 55, '09:30:00', '17:30:00', '2022-02-25', 1, 'Ogdenville'),
(11, 26, 'LIC-1011', 11, 130, '10:30:00', '18:30:00', '2022-03-30', 1, 'North Haverbrook'),
(12, 27, 'LIC-1012', 12, 75, '11:30:00', '19:30:00', '2022-04-05', 1, 'Brockway'),
(13, 28, 'LIC-1013', 13, 90, '06:30:00', '14:30:00', '2022-05-10', 1, 'Springfield'),
(14, 29, 'LIC-1014', 14, 60, '07:00:00', '15:00:00', '2022-06-15', 1, 'Shelbyville'),
(15, 30, 'LIC-1015', 15, 105, '08:00:00', '16:00:00', '2022-07-20', 1, 'Capital City');
select* from collectors
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CollectorID	UserID	LicenseNumber	CurrentVehicleID	TotalCollections	Shift_Start	Shift_End	EmploymentDate	ActiveStatus	AreaAssigned	SafetyTrainingDate
1	16	LJC-1001	1	120	07:00:00	15:00:00	2021-05-10	1	Springfield	2021
2	17	LJC-1002	2	95	08:00:00	16:00:00	2021-06-15	1	Shelbyville	2021
3	18	LJC-1003	3	80	09:00:00	17:00:00	2021-07-20	1	Capital City	2021
4	19	LJC-1004	7	110	10:00:00	18:00:00	2021-08-25	1	Downtown Core	2021
5	20	LJC-1005	5	65	11:00:00	19:00:00	2021-09-30	1	North Haverbrook	2021
6	21	LJC-1006	6	150	06:00:00	14:00:00	2021-10-05	1	Brookway	2021
7	22	LJC-1007	7	70	12:00:00	20:00:00	2021-11-10	1	Springfield	2021

INSERT INTO Admins (AdminID, UserID, ReportsGenerated, LastLoginDate, WorkShift, ApprovalCount)

VALUES

(1, 31, 25, '2023-07-01', 'Morning (8AM-4PM)', 150),
 (2, 32, 18, '2023-07-02', 'Afternoon (12PM-8PM)', 120),
 (3, 33, 30, '2023-07-03', 'Evening (4PM-12AM)', 200),
 (4, 34, 12, '2023-07-04', 'Morning (8AM-4PM)', 80),
 (5, 35, 22, '2023-07-05', 'Night (12AM-8AM)', 90),
 (6, 36, 15, '2023-07-06', 'Afternoon (12PM-8PM)', 110),
 (7, 37, 28, '2023-07-07', 'Morning (8AM-4PM)', 170),
 (8, 38, 10, '2023-07-08', 'Evening (4PM-12AM)', 60),
 (9, 39, 20, '2023-07-09', 'Night (12AM-8AM)', 130),
 (10, 40, 35, '2023-07-10', 'Morning (8AM-4PM)', 190),
 (11, 41, 14, '2023-07-11', 'Afternoon (12PM-8PM)', 70),
 (12, 42, 26, '2023-07-12', 'Evening (4PM-12AM)', 160),
 (13, 43, 19, '2023-07-13', 'Night (12AM-8AM)', 100),
 (14, 44, 23, '2023-07-14', 'Morning (8AM-4PM)', 140),
 (15, 45, 17, '2023-07-15', 'Afternoon (12PM-8PM)', 85);

select* from Admins

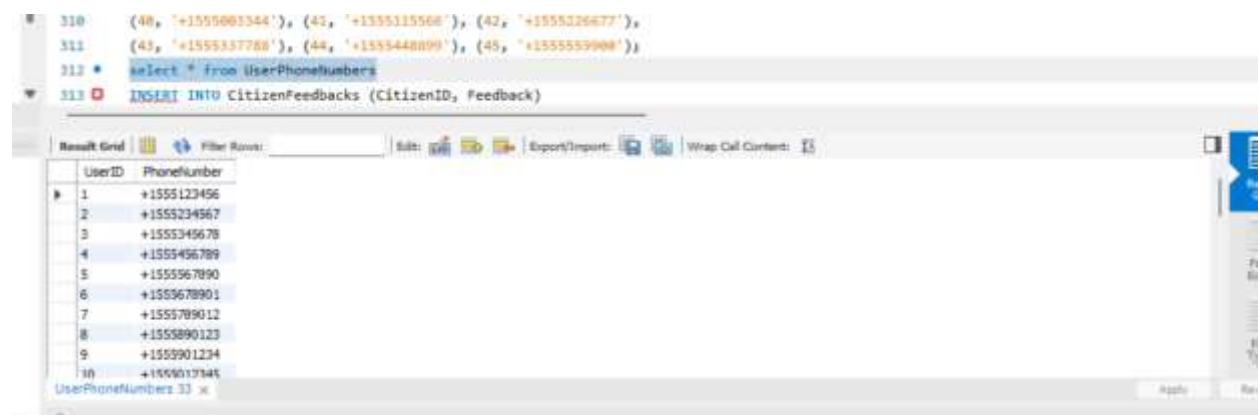
AdminID	UserID	ReportsGenerated	ApprovalCount
1	31	25	150
2	32	18	120
3	33	30	200
4	34	12	80
5	35	22	90
6	36	15	110
7	37	28	170
8	38	10	60
9	39	20	130
10	40	35	190
11	41	14	70
12	42	26	160
13	43	19	100
14	44	23	140
15	45	17	85

INSERT INTO UserPhoneNumbers (UserID, PhoneNumber)

VALUES

```
(1, '+1555123456'), (2, '+1555234567'), (3, '+1555345678'),  
(4, '+1555456789'), (5, '+1555567890'), (6, '+1555678901'),  
(7, '+1555789012'), (8, '+1555890123'), (9, '+1555901234'),  
(10, '+1555012345'), (11, '+1555112233'), (12, '+1555223344'),  
(13, '+1555334455'), (14, '+1555445566'), (15, '+1555556677'),  
(16, '+1555667788'), (17, '+1555778899'), (18, '+1555889900'),  
(19, '+1555990011'), (20, '+1555001122'), (21, '+1555113344'),  
(22, '+1555224455'), (23, '+1555335566'), (24, '+1555446677'),  
(25, '+1555557788'), (26, '+1555668899'), (27, '+1555779900'),  
(28, '+1555880011'), (29, '+1555991122'), (30, '+1555002233'),  
(31, '+1555114455'), (32, '+1555225566'), (33, '+1555336677'),  
(34, '+1555447788'), (35, '+1555558899'), (36, '+1555669900'),  
(37, '+1555770011'), (38, '+1555881122'), (39, '+1555992233'),  
(40, '+1555003344'), (41, '+1555115566'), (42, '+1555226677'),  
(43, '+1555337788'), (44, '+1555448899'), (45, '+1555559900');
```

select * from UserPhoneNumbers



INSERT INTO CitizenFeedbacks (CitizenID, Feedback)

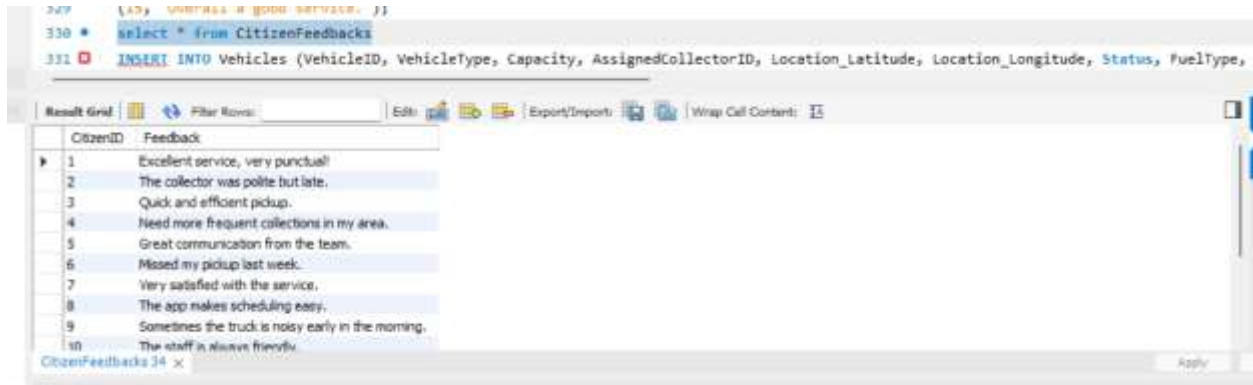
VALUES

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(1, 'Excellent service, very punctual!'),  
(2, 'The collector was polite but late.'),  
(3, 'Quick and efficient pickup.'),  
(4, 'Need more frequent collections in my area.'),  
(5, 'Great communication from the team.'),
```

```

(6, 'Missed my pickup last week.'),
(7, 'Very satisfied with the service.'),
(8, 'The app makes scheduling easy.'),
(9, 'Sometimes the truck is noisy early in the morning.'),
(10, 'The staff is always friendly.'),
(11, 'Had an issue with a missed pickup, but it was resolved.'),
(12, 'More recycling options would be great.'),
(13, 'Always on time!'),
(14, 'The collector damaged my bin once.'),
(15, 'Overall a good service.');
```

```
select * from CitizenFeedbacks
```



```
INSERT INTO Vehicles (VehicleID, VehicleType, Capacity, AssignedCollectorID, Location_Latitude,
Location_Longitude, Status, FuelType, OdometerReading)
```

```
VALUES
```

```

(1, 'Garbage Truck', 5000.00, 1, 34.052235, -118.243683, 'Available', 'Diesel', 12500),
(2, 'Recycling Van', 3000.00, 2, 34.052236, -118.243684, 'Assigned', 'Electric', 8700),
(3, 'Compactor Truck', 6000.00, 3, 34.052237, -118.243685, 'Maintenance', 'CNG', 15400),
(4, 'Pickup Truck', 2000.00, 4, 34.052238, -118.243686, 'Available', 'Gasoline', 9200),
(5, 'Dump Truck', 4500.00, 5, 34.052239, -118.243687, 'Assigned', 'Diesel', 11200),
(6, 'Electric Cart', 1000.00, 6, 34.052240, -118.243688, 'Available', 'Electric', 5600),
(7, 'Waste Hauler', 5500.00, 7, 34.052241, -118.243689, 'Assigned', 'Diesel', 14300),
(8, 'Recycling Truck', 3500.00, 8, 34.052242, -118.243690, 'Available', 'CNG', 7800),
(9, 'Garbage Truck', 5000.00, 9, 34.052243, -118.243691, 'Maintenance', 'Diesel', 16500),
(10, 'Mini Loader', 1500.00, 10, 34.052244, -118.243692, 'Available', 'Gasoline', 6400),
```

```
(11, 'Compactor Truck', 6000.00, 11, 34.052245, -118.243693, 'Assigned', 'Diesel', 13200),
(12, 'Recycling Van', 3000.00, 12, 34.052246, -118.243694, 'Available', 'Electric', 7100),
(13, 'Dump Truck', 4500.00, 13, 34.052247, -118.243695, 'Assigned', 'Diesel', 9800),
(14, 'Electric Cart', 1000.00, 14, 34.052248, -118.243696, 'Available', 'Electric', 5200),
(15, 'Waste Hauler', 5500.00, 15, 34.052249, -118.243697, 'Maintenance', 'Diesel', 14700);

select * from Vehicles
```

VehicleID	VehicleType	Capacity	AssignedCollectorID	Location_Latitude	Location_Longitude	Status	OdometerReading
1	Garbage Truck	5000.00	1	34.052235	-118.243683	Available	12500
2	Recycling Van	3000.00	2	34.052236	-118.243684	Assigned	8700
3	Compactor Truck	6000.00	3	34.052237	-118.243685	Maintenance	15400
4	Pickup Truck	2000.00	4	34.052238	-118.243686	Maintenance	9200
5	Dump Truck	4500.00	5	34.052239	-118.243687	Assigned	11200
6	Electric Cart	1000.00	6	34.052240	-118.243688	Available	5600
7	Waste Hauler	5500.00	7	34.052241	-118.243689	Assigned	14300
8	Recycling Truck	3500.00	8	34.052242	-118.243690	Available	7800
9	Garbage Truck	5000.00	9	34.052243	-118.243691	Maintenance	16500
10	Mini loader	1200.00	10	34.052244	-118.243692	Available	6400

INSERT INTO RecycleCenters

```
(CenterID, CenterName, Address_Street, Address_City, Address_ZipCode,
ContactNumber, AdminID, TotalWasteReceived, CenterCapacity, InventoryID, EstablishedDate)
```

VALUES

```
(1, 'Green Earth Recycling', '100 Eco Drive', 'Springfield', '12345', '+1555666777', 1, 50000.50, 100000.00, 1,
'2015-05-10'),
(2, 'Eco-Friendly Hub', '200 Green Lane', 'Shelbyville', '23456', '+1555777888', 2, 45000.75, 95000.00, 2,
'2016-06-15'),
(3, 'Sustainable Waste Solutions', '300 Clean Ave', 'Capital City', '34567', '+1555888999', 3, 60000.25,
110000.00, 3, '2017-07-20'),
(4, 'Planet Saver Center', '400 Recycle Blvd', 'Ogdenville', '45678', '+1555999000', 4, 35000.00, 80000.00, 4,
'2018-08-25'),
(5, 'Earth First Recycling', '500 Renew Rd', 'North Haverbrook', '56789', '+1555000111', 5, 40000.50,
90000.00, 5, '2019-09-30'),
(6, 'Green Valley Recyclers', '600 Eco Park', 'Brockway', '67890', '+1555111222', 6, 55000.75, 105000.00, 6,
'2020-10-05'),
(7, 'Pure Earth Facility', '700 Sustain St', 'Springfield', '78901', '+1555222333', 7, 48000.25, 97000.00, 7,
'2021-11-10'),
(8, 'EcoCycle Depot', '800 Green Way', 'Shelbyville', '89012', '+1555333444', 8, 52000.00, 102000.00, 8,
'2022-12-15'),
(9, 'Clean Future Center', '900 Reuse Lane', 'Capital City', '90123', '+1555444555', 9, 47000.50, 99000.00, 9,
'2023-01-20'),
```


(10, 'Nature's Recyclers', '1000 Earth Rd', 'Ogdenville', '01234', '+1555555666', 10, 43000.75, 88000.00, 10, '2020-02-25'),

(11, 'GreenTech Recycling', '1100 Bio Ave', 'North Haverbrook', '12340', '+1555666777', 11, 51000.25, 101000.00, 11, '2021-03-30'),

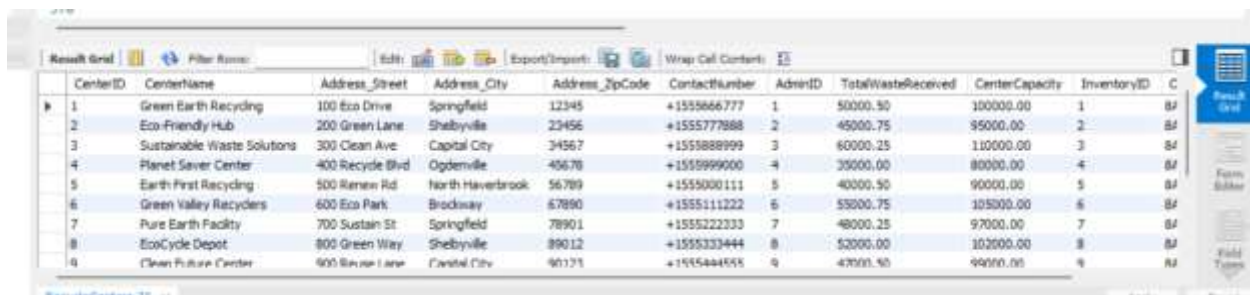
(12, 'EcoSave Station', '1200 Pure Blvd', 'Brockway', '23451', '+1555777888', 12, 39000.00, 85000.00, 12, '2022-04-05'),

(13, 'EverGreen Recyclers', '1300 Eco Drive', 'Springfield', '34562', '+1555888999', 13, 54000.50, 107000.00, 13, '2023-05-10'),

(14, 'EarthCare Facility', '1400 Green Lane', 'Shelbyville', '45673', '+1555999000', 14, 46000.75, 96000.00, 14, '2020-06-15'),

(15, 'ReNew Solutions', '1500 Clean Ave', 'Capital City', '56784', '+1555000111', 15, 49000.25, 98000.00, 15, '2021-07-20');

select * from RecycleCenters



CenterID	CenterName	Address_Street	Address_City	Address_ZipCode	ContactNumber	AdminID	TotalWasteReceived	CenterCapacity	InventoryID	C
1	Green Earth Recycling	100 Eco Drive	Springfield	12345	+1555666777	1	50000.50	100000.00	1	8/
2	Eco-Friendly Hub	200 Green Lane	Shelbyville	23456	+1555777888	2	45000.75	95000.00	2	8/
3	Sustainable Waste Solutions	300 Clean Ave	Capital City	34567	+1555888999	3	60000.25	110000.00	3	8/
4	Planet Saver Center	400 Recycle Blvd	Ogdenville	45678	+1555999000	4	35000.00	80000.00	4	8/
5	Earth First Recycling	500 Renew Rd	North Haverbrook	56789	+1555000111	5	40000.50	90000.00	5	8/
6	Green Valley Recyclers	600 Eco Park	Brockway	67890	+1555111222	6	55000.75	105000.00	6	8/
7	Pure Earth Facility	700 Sustain St	Springfield	78901	+1555222333	7	48000.25	97000.00	7	8/
8	EcoCycle Depot	800 Green Way	Shelbyville	89012	+1555333444	8	52000.00	102000.00	8	8/
9	Clean Future Center	900 Saveur Lane	Capital City	90123	+1555444555	9	47000.50	94000.00	9	8/

INSERT INTO PickupRequests (
RequestID, CitizenID, AdminID, CollectorID,
RequestDate, ApprovalStatus, TotalWeight,
Pickup_Street, Pickup_City, Pickup_ZipCode, PickupStatus
)

VALUES

(1, 1, 1, 1, '2023-05-01', 'Approved', 12.50, '123 Main St', 'Springfield', '12345', 'Collected'),

(2, 2, 2, 2, '2023-05-02', 'Approved', 8.75, '456 Oak Ave', 'Shelbyville', '23456', 'Scheduled'),

(3, 3, 3, 3, '2023-05-03', 'Pending', 15.00, '789 Pine Rd', 'Capital City', '34567', 'Scheduled'),

(4, 4, 4, 4, '2023-05-04', 'Rejected', 5.50, '101 Elm Blvd', 'Ogdenville', '45678', 'Missed'),

(5, 5, 5, 5, '2023-05-05', 'Approved', 20.25, '202 Maple Dr', 'North Haverbrook', '56789', 'Collected'),

(6, 6, 6, 6, '2023-05-06', 'Approved', 10.75, '303 Cedar Ln', 'Brockway', '67890', 'Scheduled'),

(7, 7, 7, 7, '2023-05-07', 'Pending', 18.50, '404 Birch St', 'Springfield', '78901', 'Scheduled'),

(8, 8, 8, 8, '2023-05-08', 'Approved', 7.25, '505 Spruce Ave', 'Shelbyville', '89012', 'Collected'),


```

(9, 9, 9, 9, '2023-05-09', 'Rejected', 9.00, '606 Willow Rd', 'Capital City', '90123', 'Missed'),
(10, 10, 10, 10, '2023-05-10', 'Approved', 14.75, '707 Aspen Blvd', 'Ogdenville', '01234', 'Collected'),
(11, 11, 11, 11, '2023-05-11', 'Approved', 11.50, '808 Redwood Dr', 'North Haverbrook', '12340', 'Scheduled'),
(12, 12, 12, 12, '2023-05-12', 'Pending', 16.25, '909 Sequoia Ln', 'Brockway', '23451', 'Scheduled'),
(13, 13, 13, 13, '2023-05-13', 'Approved', 6.75, '111 Fir St', 'Springfield', '34562', 'Collected'),
(14, 14, 14, 14, '2023-05-14', 'Rejected', 13.00, '222 Juniper Ave', 'Shelbyville', '45673', 'Missed'),
(15, 15, 15, 15, '2023-05-15', 'Approved', 17.50, '333 Magnolia Rd', 'Capital City', '56784', 'Collected');
select * from PickupRequests

```

RequestID	CitizenID	AdminID	CollectorID	RequestDate	ApprovalStatus	TotalWeight	Pickup_Street	Pickup_City	PickupStatus
1	1	1	1	2023-05-01	Approved	12.500	123 Main St	Springfield	Collected
2	2	2	2	2023-05-02	Approved	8.750	456 Oak Ave	Shelbyville	Scheduled
3	3	3	3	2023-05-03	Pending	15.000	789 Pine Rd	Capital City	Scheduled
4	4	4	4	2023-05-04	Rejected	5.500	101 Elm Blvd	Ogdenville	Missed
5	5	5	5	2023-05-05	Approved	20.250	202 Maple Dr	North Haverbrook	Collected
6	6	6	6	2023-05-06	Approved	10.750	303 Cedar Ln	Brockway	Scheduled
7	7	7	7	2023-05-07	Pending	18.500	404 Birch St	Springfield	Scheduled
8	8	8	8	2023-05-08	Approved	7.250	505 Spruce Ave	Shelbyville	Collected
9	9	9	9	2023-05-09	Rejected	9.000	606 Willow Rd	Capital City	Missed
10	10	10	10	2023-05-10	Approved	14.750	707 Aspen Blvd	Ogdenville	Collected

```

INSERT INTO Inventory_ItemTypes (InventoryID, ItemType)

```

```

VALUES

```

```

-- Inventory 1

```

```

(1, 'Plastic'), (1, 'Paper'), (1, 'Glass'), (1, 'Metal'),

```

```

-- Inventory 2

```

```

(2, 'Plastic'), (2, 'Paper'), (2, 'Electronics'),

```

```

-- Inventory 3

```

```

(3, 'Organic'), (3, 'Wood'), (3, 'Rubber'),

```

```

-- Inventory 4

```

```

(4, 'Hazardous'), (4, 'Construction Debris'), (4, 'E-Waste'),

```

```

-- Inventory 5

```

```

(5, 'Plastic'), (5, 'Glass'), (5, 'Metal'),

```

```

-- Inventory 6

```

```

(6, 'Textiles'), (6, 'Paper'), (6, 'Plastic'),

```

```

-- Inventory 7

```

```

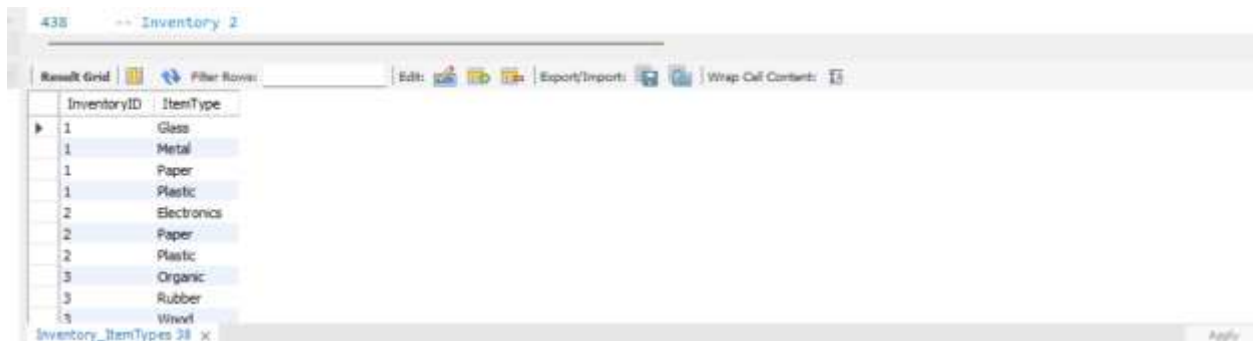
(7, 'Metal'), (7, 'Glass'), (7, 'Plastic'),

```

```

-- Inventory 8
(8, 'Organic'), (8, 'Paper'), (8, 'Wood'),
-- Inventory 9
(9, 'Plastic'), (9, 'Metal'), (9, 'Rubber'),
-- Inventory 10
(10, 'Glass'), (10, 'Paper'), (10, 'Textiles'),
-- Inventory 11
(11, 'Plastic'), (11, 'Metal'), (11, 'Electronics'),
-- Inventory 12
(12, 'Organic'), (12, 'Wood'), (12, 'Paper'),
-- Inventory 13
(13, 'Plastic'), (13, 'Glass'), (13, 'Metal'),
-- Inventory 14
(14, 'Textiles'), (14, 'Paper'), (14, 'Plastic'),
-- Inventory 15
(15, 'Metal'), (15, 'Glass'), (15, 'Plastic');
select* from Inventory_ItemTypes

```



InventoryID	ItemType
1	Glass
1	Metal
1	Paper
1	Plastic
2	Electronics
2	Paper
2	Plastic
3	Organic
3	Rubber
3	Wood

```

INSERT INTO Inventory_StockLevels (InventoryID, ItemType, Quantity)

```

```

VALUES

```

```

-- Inventory 1
(1, 'Plastic', 20000.00), (1, 'Paper', 15000.00), (1, 'Glass', 10000.00), (1, 'Metal', 5000.50),
-- Inventory 2
(2, 'Plastic', 18000.00), (2, 'Paper', 12000.75), (2, 'Electronics', 15000.00),
-- Inventory 3

```

```
(3, 'Organic', 25000.00), (3, 'Wood', 20000.25), (3, 'Rubber', 15000.00),
-- Inventory 4
(4, 'Hazardous', 10000.00), (4, 'Construction Debris', 15000.00), (4, 'E-Waste', 10000.00),
-- Inventory 5
(5, 'Plastic', 15000.00), (5, 'Glass', 12000.50), (5, 'Metal', 13000.00),
-- Inventory 6
(6, 'Textiles', 18000.00), (6, 'Paper', 22000.75), (6, 'Plastic', 15000.00),
-- Inventory 7
(7, 'Metal', 20000.00), (7, 'Glass', 15000.25), (7, 'Plastic', 13000.00),
-- Inventory 8
(8, 'Organic', 22000.00), (8, 'Paper', 18000.00), (8, 'Wood', 12000.00),
-- Inventory 9
(9, 'Plastic', 17000.00), (9, 'Metal', 15000.50), (9, 'Rubber', 15000.00),
-- Inventory 10
(10, 'Glass', 16000.00), (10, 'Paper', 14000.75), (10, 'Textiles', 13000.00),
-- Inventory 11
(11, 'Plastic', 19000.00), (11, 'Metal', 17000.00), (11, 'Electronics', 15000.25),
-- Inventory 12
(12, 'Organic', 15000.00), (12, 'Wood', 12000.00), (12, 'Paper', 12000.00),
-- Inventory 13
(13, 'Plastic', 21000.00), (13, 'Glass', 18000.50), (13, 'Metal', 15000.00),
-- Inventory 14
(14, 'Textiles', 16000.00), (14, 'Paper', 15000.75), (14, 'Plastic', 15000.00),
-- Inventory 15
(15, 'Metal', 18000.00), (15, 'Glass', 16000.00), (15, 'Plastic', 15000.00);
select* from Inventory_StockLevels
```

467 **INSERT INTO** CollectorAssignedRequests (CollectorID, RequestID)

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Contents: 15

InventoryID	ItemType	Quantity
1	Glas	Refresh data re-executing the original query
1	Metal	3000.50
1	Paper	15000.00
1	Plastic	20000.00
2	Electronics	15000.00
2	Paper	12000.75
2	Plastic	18000.00
3	Organic	25000.00
3	Rubber	15000.00
1	Wood	20000.75

Inventory_StockLevels 39 x

INSERT INTO CollectorAssignedRequests (CollectorID, RequestID)

VALUES

(1, 1), (1, 2),

(2, 3), (2, 4),

(3, 5),

(4, 6), (4, 7),

(5, 8),

(6, 9), (6, 10),

(7, 11),

(8, 12), (8, 13),

(9, 14),

(10, 15),

(11, 1),

(12, 3),

(13, 5), (13, 7),

(14, 9), (14, 11),

(15, 13);

select*from CollectorAssignedRequests

487 **INSERT INTO** AdminManagedZones (AdminID, ZoneName)

CollectorID	RequestID
1	1
1	2
2	3
2	4
3	5
4	6
4	7
5	8
6	9
6	10

CollectorAssignedRequests 40 x Apply

INSERT INTO AdminManagedZones (AdminID, ZoneName)

VALUES

(1, 'Downtown Core'),

(1, 'Financial District'),

(2, 'Northside Residential'),

(3, 'East River Zone'),

(4, 'West End Commercial'),

(5, 'Southside Industrial'),

(6, 'Central Park Area'),

(7, 'Riverside Communities'),

(8, 'Hillside Estates'),

(9, 'Uptown District'),

(10, 'Midtown Arts Quarter'),

(11, 'Lakeview Neighborhood'),

(12, 'Green Valley'),

(12, 'Green Valley West'),

(13, 'Sunset Boulevard'),

(14, 'Oceanfront Properties'),

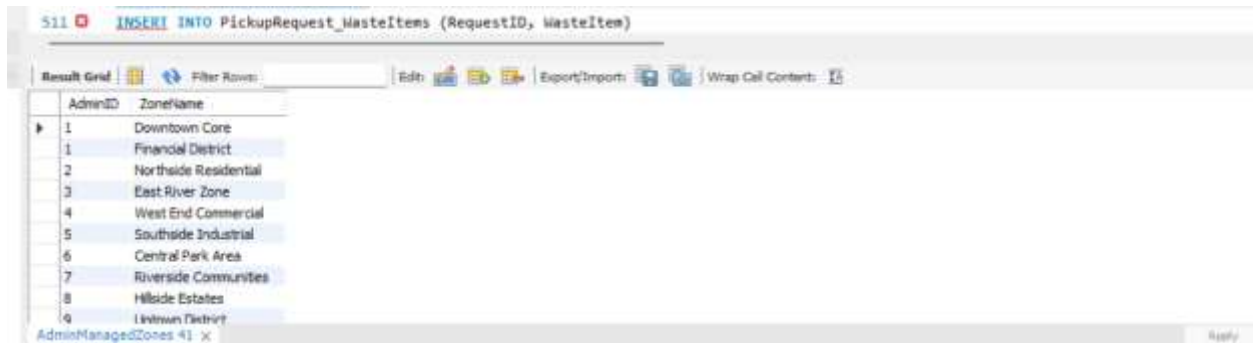
(15, 'Mountain Ridge'),

(15, 'Mountain Ridge East'),

(15, 'Mountain Ridge West'),

(9, 'Uptown Historical District');

select*from AdminManagedZones



```
INSERT INTO PickupRequest_WasteItems (RequestID, WasteItem)
```

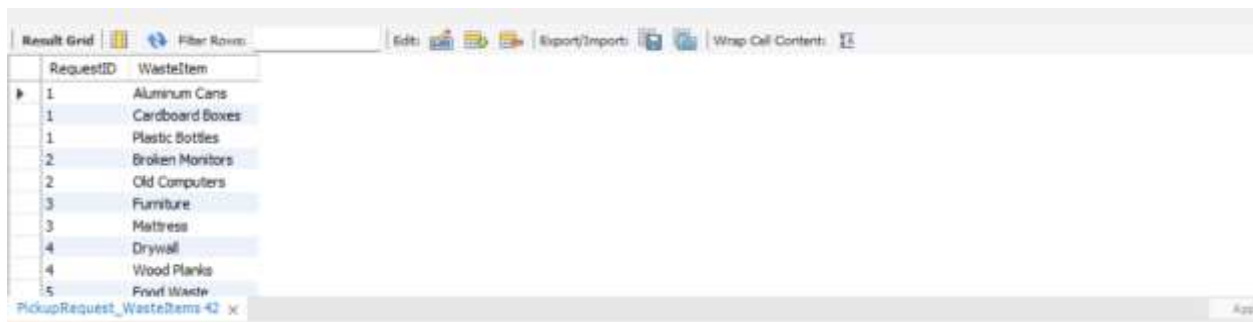
```
VALUES
```

```
(1, 'Plastic Bottles'),  
(1, 'Cardboard Boxes'),  
(1, 'Aluminum Cans'),  
(2, 'Old Computers'),  
(2, 'Broken Monitors'),  
(3, 'Furniture'),  
(3, 'Mattress'),  
(4, 'Wood Planks'),  
(4, 'Drywall'),  
(5, 'Yard Trimmings'),  
(5, 'Food Waste'),  
(6, 'Paint Cans'),  
(6, 'Batteries'),  
(7, 'Glass Bottles'),  
(7, 'Plastic Containers'),  
(8, 'Refrigerator'),  
(8, 'Washing Machine'),  
(9, 'Old Clothing'),  
(9, 'Bedding'),  
(10, 'Plastic Bags'),  
(10, 'Paper Products'),  
(11, 'Printers'),
```

```

(11, 'Cell Phones'),
(12, 'Tree Branches'),
(12, 'Grass Clippings'),
(13, 'Concrete Blocks'),
(13, 'Bricks'),
(14, 'Cleaning Chemicals'),
(14, 'Pesticides'),
(15, 'Newspapers'),
(15, 'Magazines');
select* from PickupRequest_WasteItems

```



The screenshot shows a database application window titled 'PickupRequest_WasteItems 42'. The window has a menu bar with options like 'Result Grid', 'Filter Rows', 'Edit', 'Export/Import', and 'Wrap Cell Contents'. Below the menu bar is a table with two columns: 'RequestID' and 'WasteItem'. The table contains 15 rows of data, with some rows having a blue background. The data is as follows:

RequestID	WasteItem
1	Aluminum Cans
1	Cardboard Boxes
1	Plastic Bottles
2	Broken Monitors
2	Old Computers
3	Furniture
3	Mattress
4	Drywall
4	Wood Planks
5	Food Waste

```

INSERT INTO RecycleCenter_WasteCategories (CenterID, WasteCategory)

```

```

VALUES

```

```

(1, 'Plastic'),
(1, 'Paper'),
(1, 'Glass'),
(1, 'Metal'),
(2, 'Electronics'),
(2, 'Batteries'),
(2, 'Light Bulbs'),
(3, 'Organic Waste'),
(3, 'Compost'),
(3, 'Yard Waste'),
(4, 'Wood'),
(4, 'Concrete'),

```

(4, 'Drywall'),
(5, 'Chemicals'),
(5, 'Paint'),
(5, 'Medical Waste'),
(6, 'Clothing'),
(6, 'Fabric'),
(6, 'Shoes'),
(7, 'Aluminum'),
(7, 'Copper'),
(7, 'Steel'),
(8, 'PET Plastic'),
(8, 'HDPE Plastic'),
(8, 'PVC Plastic'),
(9, 'Clear Glass'),
(9, 'Brown Glass'),
(9, 'Green Glass'),
(10, 'Cardboard'),
(10, 'Newspaper'),
(10, 'Office Paper'),
(11, 'Plastic'),
(11, 'Metal'),
(11, 'Glass'),

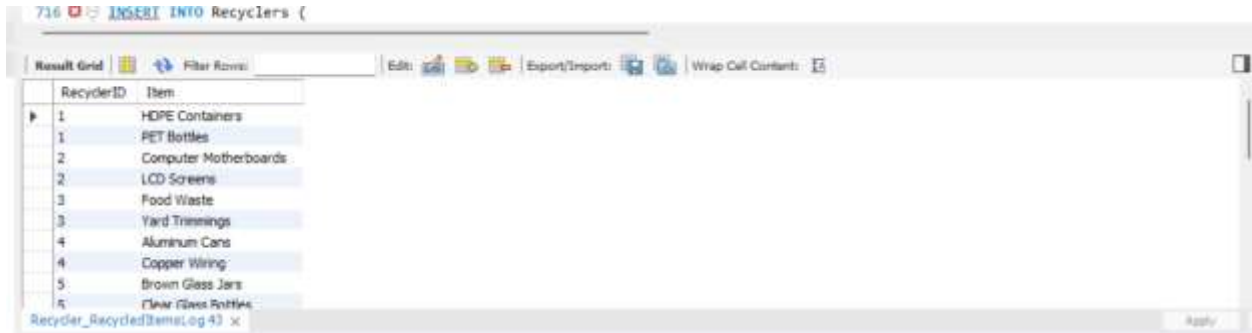
(12, 'Furniture'),
(12, 'Appliances'),
(12, 'Mattresses'),
(13, 'Motor Oil'),
(13, 'Tires'),
(13, 'Car Batteries'),
(14, 'Plastic Bags'),
(14, 'Plastic Film'),


```
(14, 'Bubble Wrap'),  
(15, 'Plastic'),  
(15, 'Paper'),  
(15, 'Glass'),  
(15, 'Metal'),  
(15, 'Electronics');  
select*from RecycleCenter_WasteCategories  
INSERT INTO Recycler_RecycledItemsLog (RecyclerID, Item)  
VALUES  
(1, 'PET Bottles'),  
(1, 'HDPE Containers'),  
(2, 'Computer Motherboards'),  
(2, 'LCD Screens'),  
(3, 'Food Waste'),  
(3, 'Yard Trimmings'),  
(4, 'Aluminum Cans'),  
(4, 'Copper Wiring'),  
(5, 'Clear Glass Bottles'),  
(5, 'Brown Glass Jars'),  
(6, 'Cardboard Boxes'),  
(6, 'Office Paper'),  
  
(7, 'Cotton Clothing'),  
(7, 'Polyester Fabric'),  
(8, 'Lead-Acid Batteries'),  
(8, 'Fluorescent Tubes'),  
(9, 'Concrete Rubble'),  
(9, 'Wood Planks'),  
(10, 'Mixed Plastics'),  
(10, 'Aluminum Foil'),  
(11, 'Plastic Bags'),
```

```

(11, 'Bubble Wrap'),
(12, 'Smartphones'),
(12, 'Tablets'),
(13, 'Motor Oil'),
(13, 'Tires'),
(14, 'Waxed Cardboard'),
(14, 'Thermal Receipts'),
(15, 'Mixed Recycling'),
(15, 'E-Waste');
select* from Recycler_RecycledItemsLog

```



RecyclerID	Item
1	HDPE Containers
1	PET Bottles
2	Computer Motherboards
2	LCD Screens
3	Food Waste
3	Yard Trimmings
4	Aluminum Cans
4	Copper Wiring
5	Brown Glass Jars
5	Clear Glass Bottles

```

INSERT INTO Recyclers (
RecyclerID, CenterID, FirstName, LastName, Email,
Phone, Specialization, Shift
)
VALUES
-- Center 1 (General Recycling)
(1, 1, 'James', 'Wilson', 'james.wilson@recycle1.com', '+15551112222', 'Plastic Sorting', 'Morning (6AM-2PM)'),
(2, 1, 'Sarah', 'Johnson', 'sarah.johnson@recycle1.com', '+15552223333', 'Glass Processing', 'Afternoon (2PM-10PM)'),

-- Center 2 (E-Waste)
(3, 2, 'Michael', 'Brown', 'michael.brown@ewaste.com', '+15553334444', 'Electronics Dismantling', 'Morning (7AM-3PM)'),

```

(4, 2, 'Emily', 'Davis', 'emily.davis@ewaste.com', '+15554445555', 'Battery Recycling', 'Evening (3PM-11PM)'),

-- Center 3 (Organic)

(5, 3, 'David', 'Miller', 'david.miller@organic.com', '+15555556666', 'Composting', 'Day (8AM-4PM)'),

(6, 3, 'Jessica', 'Taylor', 'jessica.taylor@organic.com', '+15556667777', 'Biowaste Processing', 'Swing (12PM-8PM)'),

-- Center 4 (Construction)

(7, 4, 'Robert', 'Anderson', 'robert.anderson@constr.com', '+15557778888', 'Wood Recycling', 'Morning (6AM-2PM)'),

(8, 4, 'Amanda', 'Thomas', 'amanda.thomas@constr.com', '+15558889999', 'Concrete Crushing', 'Afternoon (2PM-10PM)'),

-- Center 5 (Hazardous)

(9, 5, 'Daniel', 'Jackson', 'daniel.jackson@hazard.com', '+15559990000', 'Chemical Neutralization', 'Day (9AM-5PM)'),

(10, 5, 'Jennifer', 'White', 'jennifer.white@hazard.com', '+15550001111', 'Paint Processing', 'Evening (4PM-12AM)'),

-- Center 6 (Textile)

(11, 6, 'Christopher', 'Harris', 'christopher.harris@textile.com', '+15551112222', 'Fabric Sorting', 'Morning (7AM-3PM)'),

(12, 6, 'Elizabeth', 'Martin', 'elizabeth.martin@textile.com', '+15552223333', 'Clothing Repair', 'Afternoon (3PM-11PM)'),

-- Center 7 (Metal)

(13, 7, 'Matthew', 'Garcia', 'matthew.garcia@metal.com', '+15553334444', 'Aluminum Smelting', 'Night (10PM-6AM)'),

(14, 7, 'Ashley', 'Martinez', 'ashley.martinez@metal.com', '+15554445555', 'Copper Extraction', 'Day (8AM-4PM)'),

-- Center 8 (Plastic Specialty)

(15, 8, 'Andrew', 'Robinson', 'andrew.robinson@plastic.com', '+15555556666', 'PET Processing', 'Swing (12PM-8PM)');

select*from Recyclers

The screenshot shows a SQL IDE with an SQL statement in the top pane and a table of Recyclers in the bottom pane.

SQL Statement:

```

753 INSERT INTO Reports (
754   ReportID, AdminID, GeneratedDate, TimePeriodCovered,
755   TotalPickups, TotalRecycled, IssuesFound,
756   WasteTypeBreakdown, ReviewedBy,
757   LocationStats_Latitude, LocationStats_Longitude
758 )
  
```

Recyclers Table:

RecyclerID	CenterID	FirstName	LastName	Email	Phone	Specialization	Shift
1	1	James	Wilson	james.wilson@recycle1.com	+15551112222	Plastic Sorting	Morning (6AM-2PM)
2	1	Sarah	Johnson	sarah.johnson@recycle1.com	+15552223333	Glass Processing	Afternoon (2PM-10PM)
3	2	Michael	Brown	michael.brown@ewaste.com	+15553334444	Electronics Dismantling	Morning (7AM-3PM)
4	2	Emily	Davis	emily.davis@ewaste.com	+15554445555	Battery Recycling	Evening (3PM-11PM)
5	3	David	Miller	david.miller@organic.com	+15555556666	Composting	Day (8AM-4PM)
6	3	Jessica	Taylor	jessica.taylor@organic.com	+15556667777	Biowaste Processing	Swing (12PM-8PM)
7	4	Robert	Anderson	robert.anderson@constr.com	+15557778888	Wood Recycling	Morning (6AM-2PM)
8	4	Amanda	Thomas	amanda.thomas@constr.com	+15558889999	Concrete Crushing	Afternoon (2PM-10PM)
9	5	Daniel	Jackson	daniel.jackson@hazard.com	+15559990000	Chemical Neutralization	Day (9AM-5PM)
10	5	Iennifer	White	iennifer.white@hazard.com	+15550001111	Paint Processing	Evening (4PM-12AM)

```

INSERT INTO Reports (
  ReportID, AdminID, GeneratedDate, TimePeriodCovered,
  TotalPickups, TotalRecycled, IssuesFound,
  WasteTypeBreakdown, ReviewedBy,
  LocationStats_Latitude, LocationStats_Longitude
)
  
```

VALUES

-- Monthly Reports (Admin 31-35)

```

(1, 1, '2023-06-30', 'June 2023', 450, 320,
'Increased plastic contamination in recycling stream',
'Plastic: 42%, Paper: 28%, Glass: 18%, Metal: 12%',
'John Smith', 34.052235, -118.243683),
  
```

```

(2, 2, '2023-06-30', 'June 2023', 380, 275,
'Vehicle maintenance delays in Northside',
'Plastic: 38%, Paper: 31%, Glass: 15%, Metal: 16%',
'Jane Doe', 34.052236, -118.243684),
  
```

```

(3, 3, '2023-06-30', 'June 2023', 520, 410,
'High e-waste volume requires additional processing',
  
```

'Electronics: 22%, Plastic: 35%, Other: 43%',
'Robert Brown', 34.052237, -118.243685),

(4, 4, '2023-06-30', 'June 2023', 290, 190,
'Low participation in Ogdenville recycling program',
'Plastic: 45%, Paper: 30%, Glass: 15%, Metal: 10%',
'Emily Davis', 34.052238, -118.243686),

(5, 5, '2023-06-30', 'June 2023', 410, 350,
'Successful hazardous waste collection event',
'Hazardous: 18%, Plastic: 32%, Paper: 25%, Other: 25%',
'Michael Wilson', 34.052239, -118.243687),

-- Quarterly Reports (Admin 36-40)

(6, 6, '2023-06-30', 'Q2 2023', 1250, 950,
'Need more cardboard compactors in downtown area',
'Cardboard: 28%, Plastic: 30%, Glass: 20%, Other: 22%',
'Sarah Taylor', 34.052240, -118.243688),

(7, 7, '2023-06-30', 'Q2 2023', 1420, 1100,
'Increased glass recycling after new bins installed',
'Glass: 25%, Plastic: 35%, Paper: 25%, Metal: 15%',
'David Anderson', 34.052241, -118.243689),

(8, 8, '2023-06-30', 'Q2 2023', 980, 720,
'Staff shortages affecting collection times',
'Plastic: 40%, Paper: 30%, Glass: 15%, Other: 15%',
'Jessica Thomas', 34.052242, -118.243690),

(9, 9, '2023-06-30', 'Q2 2023', 1100, 850,
'New recycling education program showing results',

'Plastic: 38%, Paper: 28%, Glass: 22%, Metal: 12%',
'Daniel Jackson', 34.052243, -118.243691),

(10, 10, '2023-06-30', 'Q2 2023', 1350, 1050,
'Improved sorting at material recovery facility',
'Plastic: 33%, Paper: 30%, Glass: 20%, Metal: 17%',
'Lisa White', 34.052244, -118.243692),

-- Special Reports (Admin 41-45)

(11, 11, '2023-06-15', 'E-Waste Initiative', 180, 150,
'Successful pilot program for small electronics',
'Electronics: 65%, Plastic: 20%, Other: 15%',
'Matthew Harris', 34.052245, -118.243693),

(12, 12, '2023-06-20', 'Organic Waste Audit', 320, 280,
'Contamination from plastic bags in compost',
'Food Waste: 60%, Yard Waste: 35%, Contaminants: 5%',
'Amanda Martin', 34.052246, -118.243694),

(13, 13, '2023-06-25', 'Hazardous Materials', 95, 80,
'Need better household hazardous waste education',
'Paint: 40%, Chemicals: 30%, Batteries: 20%, Other: 10%',
'Christopher Garcia', 34.052247, -118.243695),

(14, 14, '2023-06-10', 'Textile Recycling', 210, 175,
'Growing demand for clothing recycling bins',
'Clothing: 70%, Shoes: 15%, Other Textiles: 15%',
'Ashley Lee', 34.052248, -118.243696),

(15, 15, '2023-06-05', 'Construction Recycling', 150, 120,
'Challenges with mixed construction materials',

'Wood: 40%, Drywall: 30%, Concrete: 20%, Other: 10%',

'James Clark', 34.052249, -118.243697);

select*from Reports

ReportID	AdminID	GeneratedDate	TimePeriodCovered	TotalPickups	TotalRecycled	IssuesFound	WasteTypeBreakdown
1	1	2023-06-30	June 2023	450	320	Increased plastic contamination in recycling stream...	Plastic: 42%, Paper: 28%, Glass: 18%, Metal: 12%
2	2	2023-06-30	June 2023	380	275	Vehicle maintenance delays in Northside	Plastic: 38%, Paper: 31%, Glass: 15%, Metal: 16%
3	3	2023-06-30	June 2023	520	410	High e-waste volume requires additional processing...	Electronics: 22%, Plastic: 35%, Other: 43%
4	4	2023-06-30	June 2023	290	190	Low participation in Ogdenville recycling program	Plastic: 45%, Paper: 30%, Glass: 15%, Metal: 10%
5	5	2023-06-30	2023-06-30	410	350	Successful hazardous waste collection event	Hazardous: 18%, Plastic: 32%, Paper: 25%, Other: 25%
6	6	2023-06-30	Q2 2023	1250	950	Need more cardboard compactors in downtown area...	Cardboard: 28%, Plastic: 30%, Glass: 20%, Other: 22%
7	7	2023-06-30	Q2 2023	1420	1100	Increased glass recycling after new bins installed	Glass: 25%, Plastic: 35%, Paper: 25%, Metal: 15%
8	8	2023-06-30	Q2 2023	980	720	Staff shortages affecting collection times	Plastic: 40%, Paper: 30%, Glass: 15%, Other: 15%
9	9	2023-06-30	Q2 2023	1100	850	New recycling education program showing results	Plastic: 38%, Paper: 28%, Glass: 22%, Metal: 12%

INSERT INTO Report_TopCollectors (ReportID, CollectorID)

VALUES

-- Monthly Reports (ReportID 1-5)

(1, 1), -- Top collector for June 2023 report (Springfield)

(1, 2), -- Second best collector

(2, 3), -- Top collector for Northside June report

(2, 4),

(3, 5), -- E-waste collection leader

(3, 6),

(4, 7), -- Ogdenville top performer

(5, 8), -- Hazardous waste collection star

-- Quarterly Reports (ReportID 6-10)

(6, 9), -- Q2 top cardboard collector

(6, 10),

(7, 11), -- Glass recycling champion

(7, 12),

(8, 13), -- High-volume collector

(9, 14), -- Education program standout

(10, 15), -- Most improved collector

```

-- Special Reports (ReportID 11-15)
(11, 1), -- E-waste initiative leader
(12, 3), -- Organic waste top collector
(13, 5), -- Hazardous materials specialist
(14, 7), -- Textile recycling star
(15, 9); -- Construction waste leader
select * from Report_TopCollectors

INSERT INTO Collector_Deliveries (CollectorID, CenterID, DeliveryDate)
VALUES

-- Week 1 Deliveries (Collectors 1-5)
(1, 1, '2023-06-05'), -- Collector 1 to General Recycling Center
(2, 2, '2023-06-06'), -- Collector 2 to E-Waste Center
(3, 3, '2023-06-07'), -- Collector 3 to Organic Center
(4, 4, '2023-06-08'), -- Collector 4 to Construction Center
(5, 5, '2023-06-09'), -- Collector 5 to Hazardous Center


-- Week 2 Deliveries (Collectors 6-10)
(6, 6, '2023-06-12'), -- Collector 6 to Textile Center
(7, 7, '2023-06-13'), -- Collector 7 to Metal Center
(8, 8, '2023-06-14'), -- Collector 8 to Plastic Center
(9, 1, '2023-06-15'), -- Collector 9 to General Recycling (mixed load)
(10, 2, '2023-06-16'), -- Collector 10 to E-Waste Center


-- Week 3 Deliveries (Collectors 11-15)
(11, 3, '2023-06-19'), -- Collector 11 to Organic Center
(12, 4, '2023-06-20'), -- Collector 12 to Construction Center
(13, 5, '2023-06-21'), -- Collector 13 to Hazardous Center
(14, 6, '2023-06-22'), -- Collector 14 to Textile Center
(15, 7, '2023-06-23'); -- Collector 15 to Metal Center

select*from Collector_Deliveries

```


ALTER TABLE Users

ADD LastLogin DATETIME;

ALTER TABLE Users

ADD AccountStatus VARCHAR(20) DEFAULT 'Active';

select*from Users

ALTER TABLE Citizens

DROP COLUMN PreferredPickupTime;

select*from Citizens

ALTER TABLE PickupRequests

MODIFY TotalWeight DECIMAL(10,3);

ALTER TABLE PickupRequests

DROP COLUMN Pickup_ZipCode;

select*from PickupRequests

ALTER TABLE Collectors

ADD SafetyTrainingDate DATE

select*from Collectors

ALTER TABLE RecycleCenters

ADD OperatingHours VARCHAR(50) DEFAULT '8AM-5PM'

select*from RecycleCenters

ALTER TABLE Inventory

DROP COLUMN InventoryManagerID

select*from Inventory

ALTER TABLE Admins

DROP COLUMN LastLoginDate

ALTER TABLE Admins

DROP COLUMN WorkShift

select*from Admins

ALTER TABLE Vehicles

DROP COLUMN FuelType

select*from Vehicles

ALTER TABLE RecycleCenters

DROP COLUMN EstablishedDate

select*from RecycleCenters

ALTER TABLE Citizens

DROP COLUMN Location_Latitude

select*from Citizens

UPDATE Collectors

SET CurrentVehicleID = 7,

AreaAssigned = 'Downtown Core'

WHERE CollectorID = 4;

DELETE FROM Collectors

WHERE ActiveStatus = 0;

select *from Collectors

UPDATE PickupRequests

SET PickupStatus = 'Collected',

ApprovalStatus = 'Approved'

WHERE RequestID IN (5, 8, 12)

AND PickupStatus = 'Scheduled'

select * from PickupRequests

UPDATE Vehicles

SET Status = 'Maintenance'

WHERE VehicleID = 4

select *from Vehicles

SELECT RequestID, Pickup_City, PickupStatus

FROM PickupRequests
WHERE ApprovalStatus = 'Pending'

SELECT FirstName, LastName
FROM Users

SELECT ItemType, Quantity
FROM Inventory_StockLevels

WHERE Quantity < 50

UPDATE Collectors

SET TotalCollections = TotalCollections + 1

WHERE CollectorID = 7

select * from Collectors

SELECT * FROM Collectors

WHERE TotalCollections > 100

SELECT * FROM Vehicles

WHERE Capacity < 3000

SELECT * FROM Citizens

WHERE NumberOfRequestsMade = 5

SELECT * FROM PickupRequests

WHERE TotalWeight != 10.00

SELECT * FROM Collectors

WHERE ActiveStatus = 1 AND AreaAssigned = 'Northside'

SELECT * FROM PickupRequests

WHERE Pickup_City = 'Springfield' OR Pickup_City = 'Shelbyville'

SELECT * FROM Users

WHERE UserType = 'citizen' AND Address_City != 'Capital City'

SELECT * FROM Users

WHERE Email LIKE '%@gmail.com'

SELECT * FROM RecycleCenters

WHERE CenterName LIKE 'Green%'

SELECT * FROM UserPhoneNumbers

WHERE PhoneNumber LIKE '%555%'

SELECT * FROM PickupRequests

WHERE Pickup_City IN ('Springfield', 'Shelbyville', 'Capital City')

SELECT * FROM PickupRequests

WHERE RequestDate BETWEEN '2023-06-01' AND '2023-06-30'

SELECT * FROM PickupRequests

WHERE TotalWeight > 15 AND Pickup_City = 'Springfield'

SELECT * FROM PickupRequests

WHERE PickupStatus = 'Scheduled' AND ApprovalStatus = 'Approved'

drop table Collector_Deliveries

drop table Report_TopCollectors

drop table Reports

drop table UserPhoneNumbers

drop table CitizenFeedbacks